between electricity and magnetism. Soon various theories of electromagnetism, notable ones being from Andre Marie Ampere, etc., were in circulation.

Yet the existence of electromagnetic waves was not even imagined until the 1850s, when Maxwell published his theories of electromagnetism. In a paper named "A dynamical theory of the electromagnetic field", Maxwell published his views regarding the existence of electromagnetic waves. He also summarised all that was known about electromagnetism at that time into the four famous Maxwell equations. Much of the groundwork to his beliefs had been laid by Michael Faraday, who established concepts such as electromagnetic induction, and theorised that electric and magnetic fields extend beyond conductors into the space around them.

In the 1870s and the 1880s, a volley of patents were filed in the United States for devices that could transmit and receive electromagnetic waves. While many of these took huge leaps of imagination, some were quite close to the modern idea of radio. Notable among these attempts, were some by the famed inventor Thomas Alva Edison.

The first major milestone towards wireless transmissions was achieved between 1886 and 1888 by Heinrich Rudolf Hertz, after whom the SI unit of frequency is today named. Hertz demonstrated the transmission and reception of radio signals and was the first person to do so. He also discovered that Maxwell's equations could be reformulated to form a differential equation, from which could be derived the wave equation.

Nikola Tesla, Inventor and Genius.

Yet the true birth of modern wireless can be seen in Nikola Tesla's famous article "the true wireless". He soon demonstrated the transmission and reception of radio waves, and then gave a lecture on the principles of wireless